

Ethics in Data Science (22UAD915) - Question Bank

UNIT I: INTRODUCTION AND OVERVIEW

Why Ethics for Data Scientists

1. Define ethics. (L1)
2. What is data science ethics? (L1)
3. List any two reasons why ethics is important for data scientists. (L1)
4. State the meaning of "moral responsibility" in data science. (L1)
5. Mention two consequences of ignoring ethics in data-driven decisions. (L1)
6. Explain briefly why data scientists need ethical guidelines despite technical expertise. (L2)
7. Summarize the role of ethics in preventing data misuse. (L2)

Topic: Difference Between Ethics and Law

8. Define law. (L1)
9. List two differences between ethics and law. (L1)
10. State whether all illegal actions are unethical. (L1)
11. Give one example of an act that is legal but unethical. (L1)
12. Differentiate between ethics and law based on enforcement. (L2)
13. Explain how ethics can go beyond legal requirements. (L2)

Topic: Working with Cases

14. What is a case study? (L1)
15. Mention the purpose of using case studies in ethics education. (L1)
16. List two benefits of analyzing real-world cases in data science ethics. (L2)
17. Explain how case studies help identify ethical dilemmas. (L2)

Topic: Need for Studying Cases

18. State one reason why studying cases is necessary in data ethics. (L1)
19. Identify two skills developed by studying ethical case studies. (L1)
20. Summarize how case-based learning improves decision-making in data science. (L2)

Scenario-Based Questions

21. A company legally collects user data through consent forms but sells it without users truly understanding the terms. Is this ethical or unethical? Justify in one line. (L4)
22. A data scientist follows all company laws but ignores the social impact of a biased model. Identify the ethical gap in one line. (L4)
23. Two data scientists disagree on whether an action is right, though it is legal. What does this indicate about ethics vs. law? (L3)
24. A new employee is asked to analyze a controversial case study before starting a data project. What is the purpose of this exercise? (L3)
25. A firm faces public backlash for a "legal" data practice. Explain briefly why legality alone was not sufficient here. (L4)

8-Mark Questions

1. Explain the concept of ethics and discuss its significance for data scientists. (L2)

2. Discuss why studying ethics is essential in the field of data science with suitable examples. **(L2)**
3. Differentiate between ethics and law, highlighting their scope and enforcement mechanisms. **(L2)**
4. Explain with examples how an action can be legal but unethical. **(L3)**
5. Discuss the role of case studies in understanding ethical dilemmas in data science. **(L2)**
6. Analyze why case-based learning is more effective than theoretical study alone in data ethics. **(L4)**
7. Explain the relationship between law, morality, and professional conduct in data science. **(L2)**
8. Apply the concept of "ethics vs. law" to justify a situation where legal compliance does not guarantee ethical behavior. **(L3)**
9. Discuss the consequences faced by organizations that ignore ethical considerations despite being legally compliant. **(L4)**
10. Analyze the need for a structured framework to study ethical cases in data science. **(L4)**

16-Mark Questions

1. Explain in detail the concept of ethics, its importance, and its relevance to data science professionals. **(L2)**
2. Discuss comprehensively the differences between ethics and law, supporting your answer with real-world examples from data science. **(L2)**
3. Analyze the need for studying ethical case studies in data science and explain how they help build professional judgment. **(L4)**
4. Explain, with detailed examples, situations where legal actions in data handling have led to ethical controversies. **(L3)**
5. Discuss the evolution of ethics in the context of technology and data-driven decision-making. **(L2)**
6. Analyze the challenges data scientists face in balancing legal compliance with ethical responsibility. **(L4)**
7. Explain the importance of case study analysis as a tool for ethical training in data science education. **(L2)**
8. Apply ethical and legal frameworks to critically examine a real or hypothetical data misuse incident. **(L3)**
9. Discuss the various factors that differentiate a legally sound decision from an ethically sound decision. **(L4)**
10. Analyze how the absence of ethical awareness among data professionals can lead to long-term organizational and societal harm. **(L4)**

Scenario-Based Questions – 8 Marks

11. A start-up legally scrapes publicly available social media data to build a facial recognition tool but does not inform users. Analyze the ethical implications of this action, even though no law was broken. **(L4)**
12. A data analyst is instructed by management to follow only the minimum legal requirements while designing a data collection policy. Discuss whether this approach is ethically sufficient, with justification. **(L4)**
13. Two employees at a company disagree on whether an algorithm's outcome is "right," even though it complies with all regulations. Analyze the ethics-vs-law conflict illustrated in this situation. **(L4)**
14. A company introduces a mandatory case-study training program for new data science hires before allowing them to handle live projects. Explain the purpose and benefits of this practice. **(L3)**

15. A firm faces severe public criticism for a data practice that was fully legal. Analyze why legality alone failed to protect the company's reputation, and suggest what additional considerations were missing. (L4)

Scenario-Based Questions – 16 Marks

16. A multinational company operates in a country where data privacy laws are minimal, allowing it to legally collect and use customer data without restriction. However, public outrage follows after journalists expose the scale of data collection. Analyze this scenario in detail using the concepts of ethics vs. law, and discuss what ethical obligations existed beyond legal compliance. (L4)
17. A data science team is given a case study of a company that manipulated user behavior through legal but ethically dubious data practices. Discuss in detail how studying this case would help the team avoid similar issues in their own projects, and explain the broader need for case-based ethical education. (L4)
18. A junior data scientist notices that a widely used dataset in their organization was legally obtained but originally collected without proper user consent decades ago. Analyze the ethical dilemma faced by the data scientist and discuss the difference between legal ownership and ethical use of data. (L4)
19. An organization repeatedly cites "we did not break any law" as a defense whenever ethical concerns are raised about its data practices. Critically analyze this stance and discuss, with examples, why ethics must be treated as a standard independent of and often exceeding legal requirements. (L4)
20. A newly hired data science graduate is asked to justify, using both ethical and legal reasoning, whether a controversial industry practice (e.g., selling anonymized user data) should continue. Construct a detailed argument analyzing this practice from both perspectives and explain why studying such cases is essential for professional development. (L4)

UNIT II: CONTENT OVERVIEW AND ANALYTICS

Ethics of Autonomy

1. Define autonomy in the context of data ethics. (L1)
2. State the meaning of "informed consent." (L1)
3. List two elements required for user autonomy in data collection. (L1)
4. Explain how autonomy relates to user control over personal data. (L2)
5. Summarize why respecting autonomy is important in data analytics. (L2)

Professional Role Responsibilities

6. Define professional responsibility. (L1)
7. List two responsibilities of a data professional. (L1)
8. Mention one duty a data scientist owes to the public. (L1)
9. Explain the role of accountability in professional data science work. (L2)

Corporate Responsibility

10. Define corporate responsibility. (L1)
11. List two obligations of a company handling user data. (L1)
12. State one example of corporate social responsibility in data usage. (L1)
13. Explain briefly how corporate responsibility differs from individual responsibility. (L2)

Formal Codes of Ethics

14. Define a code of ethics. (L1)
15. Name one organization that has a formal data ethics code. (L1)
16. List two purposes of formal codes of ethics. (L1)
17. Explain how formal codes of ethics guide professional behavior. (L2)

Big Data and Privacy

18. Define big data. (L1)
19. Define privacy in the context of big data. (L1)
20. List two privacy risks associated with big data. (L1)
21. Explain how big data collection can threaten individual privacy. (L2)

Facebook Case Study

22. Name the case study discussed regarding big data misuse. (L1)
23. State the main ethical issue in the Facebook case study. (L1)
24. List two consequences faced by Facebook due to the data misuse case. (L1)
25. Summarize the key lesson learned from the Facebook case study. (L2)

Scenario-Based Questions (2 Marks)

26. A user unknowingly agrees to share personal data through a long, complex terms-and-conditions page. Which ethical principle is violated here? (L3)
27. An employee at a tech firm identifies a privacy flaw but is told to stay silent. Which professional responsibility is being compromised? (L3)
28. A company uses user data for purposes not stated during collection, similar to the Facebook case. Name the ethical issue involved. (L3)
29. An organization publishes a formal code of ethics but does not enforce it internally. What problem does this create? (L4)
30. A firm claims corporate responsibility while secretly selling user data to third parties. Identify the ethical contradiction. (L4)

8-Mark Questions

1. Explain the concept of autonomy in data ethics and its importance in user data collection. **(L2)**
2. Discuss the significance of informed consent in protecting user autonomy. **(L2)**
3. Explain the professional role responsibilities of a data scientist toward the public and the organization. **(L2)**
4. Discuss the concept of corporate responsibility with respect to data handling practices. **(L2)**
5. Explain the purpose and importance of formal codes of ethics in the data science profession. **(L2)**
6. Analyze the privacy risks associated with the use of big data. **(L4)**
7. Discuss the Facebook case study and the key ethical lessons it teaches about data privacy. **(L2)**
8. Apply the principle of corporate responsibility to evaluate how organizations should handle large-scale user data. **(L3)**
9. Analyze the relationship between big data collection practices and the erosion of individual privacy. **(L4)**
10. Discuss how formal codes of ethics help resolve conflicts between corporate interests and individual rights. **(L4)**

16-Mark Questions

1. Explain in detail the concept of autonomy and informed consent, and discuss their role in ethical data collection practices. **(L2)**
2. Discuss comprehensively the professional and corporate responsibilities of individuals and organizations handling user data. **(L2)**
3. Analyze the Facebook–Cambridge Analytica case study in detail and discuss the ethical violations involved along with their consequences. **(L4)**
4. Explain the importance of formal codes of ethics in the data science industry and discuss how they are implemented and enforced. **(L2)**
5. Analyze the privacy challenges posed by big data analytics and discuss possible measures to mitigate these risks. **(L4)**
6. Discuss in detail the difference between individual professional responsibility and organizational corporate responsibility in data ethics. **(L4)**
7. Apply the concept of ethics of autonomy to critically evaluate modern data collection practices used by technology companies. **(L3)**
8. Analyze how the lack of formal ethical codes contributed to major data privacy scandals in the technology industry. **(L4)**
9. Discuss the balance between big data innovation and individual privacy protection, using relevant case examples. **(L4)**
10. Analyze the broader societal impact of corporate negligence in data privacy, using the Facebook case study as a reference point. **(L4)**

Scenario-Based Questions – 8 Marks

11. A social media platform allows third-party apps to access user data along with data of the user's friends without explicit consent, similar to the Facebook case. Analyze the ethical violations involved. **(L4)**
12. An employee discovers that their organization is not following its own published code of ethics regarding data handling. Discuss the ethical responsibilities of the employee in this situation. **(L4)**
13. A company collects massive amounts of user data under the label of "improving services" but uses it for undisclosed purposes. Analyze this practice in terms of autonomy and informed consent. **(L4)**
14. A data scientist is asked to bypass certain privacy safeguards to speed up a big data analytics project. Discuss the professional responsibilities the data scientist must uphold in this situation. **(L3)**
15. A company claims to practice strong corporate responsibility but has no formal code of ethics governing its data practices. Analyze the gap between claimed and actual responsibility. **(L4)**

Scenario-Based Questions – 16 Marks

16. A popular social networking company allowed a third-party research firm to harvest personal data from millions of users without adequate consent, which was later used for political profiling. Analyze this scenario in detail with reference to the Facebook–Cambridge Analytica case, discussing violations of autonomy, corporate responsibility, and the absence of enforced ethical codes. **(L4)**
17. An organization collects extensive behavioral data through its big data analytics platform, citing business growth as justification. Employees raise concerns internally, but no formal code of ethics exists to guide decision-making. Analyze the ethical gaps in this scenario and discuss what specific codes of ethics and professional responsibilities should have been in place. **(L4)**
18. A data analytics firm markets itself as being highly privacy-conscious, yet its terms of service allow extensive data sharing with unnamed "partners." Critically analyze this scenario in

terms of informed consent and autonomy, and discuss the ethical obligations the firm has failed to meet. **(L4)**

19. Following a major data breach, an organization admits it had no formal code of ethics or clear corporate responsibility framework for handling user data. Analyze the case in detail, discussing how the absence of such structures contributed to the breach and what remedial actions should be taken. **(L4)**
20. A data scientist working on a big data project uncovers that user consent forms were intentionally written in confusing legal language to obscure how personal data would be used. Analyze this scenario in the context of autonomy, professional responsibility, and corporate ethics, and discuss the broader implications for the data science profession. **(L4)**

UNIT III: TARGET, MARKETING AND FAIRNESS IN ALGORITHMS

Ethics of Privacy

1. Define ethics of privacy. (L1)
2. List two aspects of personal data considered private. (L1)
3. State one right users have regarding their private data. (L1)
4. Explain the importance of privacy ethics in data-driven marketing. (L2)

Obligations for Organizations vs. Individuals

5. Define "obligation" in the context of data ethics. (L1)
6. List one obligation of an organization toward its users. (L1)
7. List one obligation of an individual toward data they handle. (L1)
8. Differentiate briefly between organizational and individual data obligations. (L2)
9. Explain why organizations have greater accountability than individuals in data handling. (L2)

Justice and Discrimination

9. Define discrimination in the context of algorithms. (L1)
10. State the meaning of justice in data ethics. (L1)
11. List two types of discrimination that can occur through data systems. (L1)
12. Mention one real-world example of algorithmic discrimination. (L1)
13. Explain how biased data can lead to discriminatory outcomes. (L2)

Fairness in Algorithms

14. Define fairness in algorithms. (L1)
15. List two methods to achieve fairness in machine learning models. (L1)
16. State one factor that can cause unfairness in algorithmic decisions. (L1)
17. Explain how fairness in algorithms impacts marketing and targeting practices. (L2)
18. Summarize why fairness testing is necessary before deploying an algorithm. (L2)

Scenario-Based Questions

19. An online store shows different prices to users based on their location and browsing history. Which ethical issue does this represent? (L3)
20. A loan approval algorithm rejects applicants from a particular zip code more often than others. Identify the ethical concern. (L4)
21. A company collects health-related data for targeted ads without explicit user consent. Which principle is being violated? (L3)

22. An organization blames an individual employee for a data privacy breach caused by weak company policy. Explain the obligation mismatch here. (L4)
23. A hiring algorithm consistently favors one gender due to biased training data. Name the ethical issue and the underlying cause. (L4)

8-Mark Questions

1. Explain the concept of ethics of privacy and its relevance to targeted marketing. (L2)
2. Discuss the obligations of organizations versus individuals in protecting data privacy. (L2)
3. Explain the concept of justice in the context of algorithmic decision-making. (L2)
4. Discuss how discrimination can occur through biased algorithms, with examples. (L2)
5. Explain the concept of fairness in algorithms and its importance in automated decision systems. (L2)
6. Analyze the causes of algorithmic bias and its impact on marginalized groups. (L4)
7. Discuss the methods used to achieve fairness in machine learning models. (L2)
8. Apply the principles of privacy ethics to evaluate targeted advertising practices. (L3)
9. Analyze the difference between organizational and individual obligations when a data-driven decision causes harm. (L4)
10. Discuss the relationship between fairness, justice, and discrimination in algorithmic systems. (L4)

16-Mark Questions

1. Explain in detail the ethics of privacy and discuss its application to target marketing and personalized advertising. (L2)
2. Discuss comprehensively the obligations of organizations versus individuals in ensuring ethical data practices. (L2)
3. Analyze the concept of justice and discrimination in algorithmic systems, using real-world examples of biased outcomes. (L4)
4. Explain the concept of fairness in algorithms and discuss various techniques used to detect and mitigate algorithmic bias. (L2)
5. Analyze how racial, gender, and socioeconomic biases can be embedded in data and subsequently reflected in algorithmic decisions. (L4)
6. Discuss in detail the ethical obligations of organizations in preventing discriminatory outcomes in automated systems. (L4)
7. Apply fairness principles to critically evaluate a hiring or lending algorithm and discuss the improvements needed. (L3)
8. Analyze the tension between effective target marketing and the ethical right to privacy of individuals. (L4)
9. Discuss the role of transparency and accountability in achieving fairness and justice in algorithmic decision-making. (L4)
10. Analyze the broader societal consequences of unfair and discriminatory algorithms in critical sectors such as finance, employment, and healthcare. (L4)

Scenario-Based Questions – 8 Marks

11. An e-commerce platform charges different prices to different users based on their browsing history and location. Analyze the ethical issues related to privacy and fairness in this practice. (L4)
12. A loan approval algorithm consistently denies applications from a specific neighborhood, disproportionately affecting a particular community. Analyze the type of discrimination occurring and its ethical implications. (L4)

13. A marketing company profiles users based on sensitive attributes like health conditions to deliver targeted ads without explicit consent. Discuss the privacy and fairness violations in this scenario. **(L4)**
14. An organization blames an individual employee for a discriminatory outcome produced by a flawed algorithm, while the underlying training data itself was biased. Analyze the mismatch in obligations described here. **(L4)**
15. A recruitment algorithm systematically ranks resumes from one gender higher than another due to historical hiring data used for training. Analyze the root cause and ethical concern illustrated in this scenario. **(L4)**

Scenario-Based Questions – 16 Marks

16. An online retail company uses a dynamic pricing algorithm that charges higher prices to users from wealthier zip codes and lower prices to others, based on inferred purchasing power from browsing data. Analyze this scenario in detail with respect to privacy ethics, fairness, and potential discriminatory impact, and discuss what safeguards should have been implemented. **(L4)**
17. A financial institution's AI-based loan approval system is found to reject a significantly higher percentage of applicants from certain ethnic communities compared to others, even when income and credit scores are similar. Analyze this case in depth, discussing the concepts of justice, discrimination, and algorithmic fairness, and propose measures to correct the bias. **(L4)**
18. A digital advertising company builds detailed psychological and behavioral profiles of users, including inferred sensitive information such as political views and health status, to sell highly targeted ads. Critically analyze this scenario in terms of privacy obligations of organizations versus individual rights, and discuss the ethical boundaries that should govern such practices. **(L4)**
19. A company's recruitment AI is exposed for consistently disadvantaging older candidates and candidates from certain educational backgrounds due to biases in its historical training data. Analyze this scenario comprehensively, discussing the sources of algorithmic bias, the resulting injustice, and the organizational obligations to ensure fairness. **(L4)**
20. Following public scrutiny, a company discovers that its target marketing algorithm has been unintentionally excluding certain demographic groups from seeing job and housing advertisements, a practice with potential legal and ethical consequences. Analyze this scenario in detail, discussing fairness in algorithms, justice, discrimination, and the shared obligations of the organization and its data science team in preventing such outcomes. **(L4)**